Matrix Structural Analysis

Project Report **Prof. Aritra Chatterjee**

Group – 1

Problem Statement:

The central objective of this project is to conduct a simplified structural analysis on a complex, multi-span bridge, which has been meticulously idealized as a 2D Pin-Jointed Truss.

Bridge Summary:

The bridge considered here is the Northfield Schell Bridge. It's a historic steel cantilever truss built in 1903 over the Connecticut River, known for its distinct three-span design and engineering elegance.

Way Of Analysis:

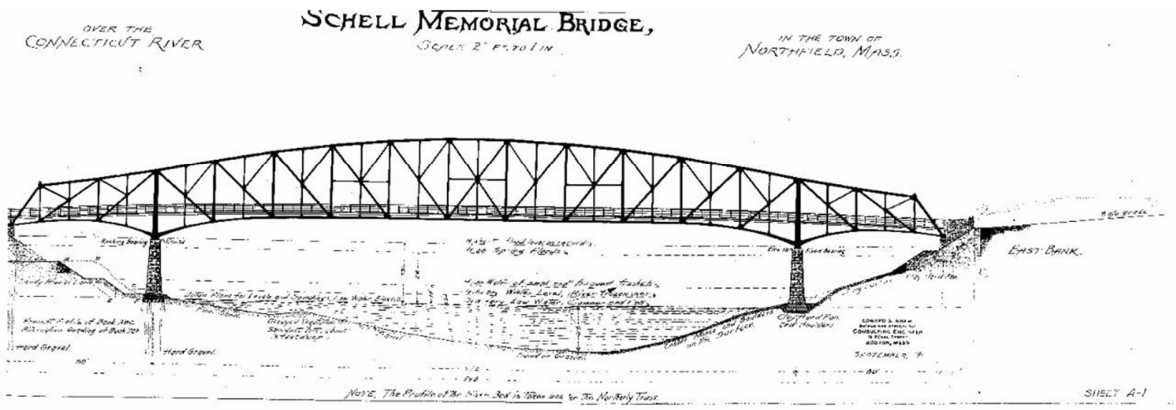
Matrix Structural Analysis: Matrix Structural Analysis (MSA) is a powerful computational technique widely used in modern structural engineering to analyze complex frameworks such as bridges, towers, and buildings. By representing structural elements and their interactions in matrix form, MSA enables engineers to efficiently compute internal forces, displacements, and reactions with high precision.

The method is grounded in the principles of linear algebra and the mechanics of materials. It divides a structure into discrete elements—such as beams or truss members—and establishes mathematical relationships between their stiffness, loads, and deformations. This matrix-based formulation allows for a systematic and scalable approach to analyzing large, indeterminate structures that would be challenging to solve using conventional hand calculations.

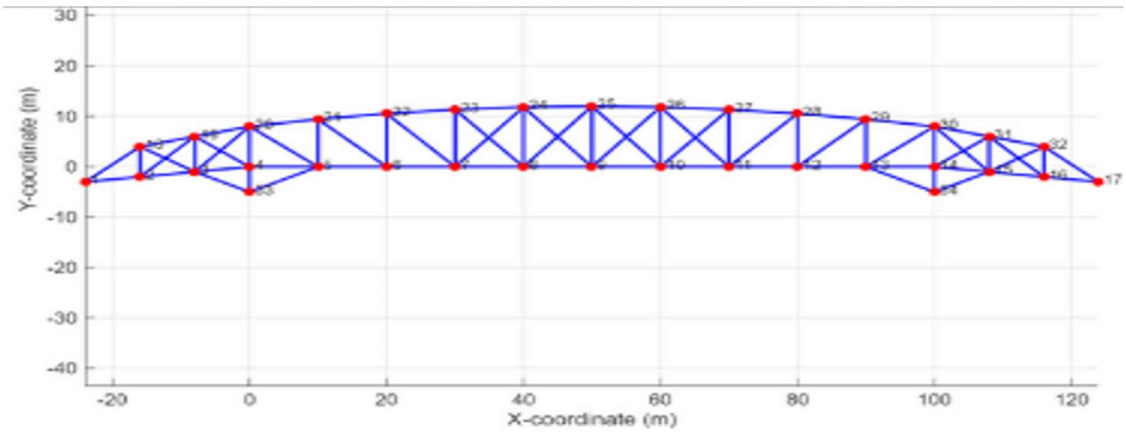
In this project, the Direct Stiffness Method (DSM)—a core component of MSA—was employed to analyze a 2D pin-jointed truss idealized from a multi-span bridge. The primary objective was to determine the axial forces, support reactions, and overall structural response under specified dead and live loads. This analytical process not only demonstrates the efficiency and accuracy of matrix methods but also deepens the understanding of how real-world truss bridges behave under various loading conditions.

Structural Idealization:
The project utilizes the concept of structural idealization, where complex, real-world details of the bridge—such as connections and secondary members—are systematically reduced. This methodological simplification allows the focus to be placed on the main structural behavior and the efficient understanding of fundamental truss concepts. The overall geometry and proportions of the truss were maintained to resemble the actual structure being analyzed.

TRUE STRUCTURE



IDEALIZED STRUCTURE



Assumptions:
1.The structure is a truss
2. The structure doesn't undergo thermal or impact(blasting) load
3. The material used in this structure is perfectly elastic
4.The Dynamics has not been considered or modeled
5.THe support at the ends is considered as pin support.

Structural Model:

1. Model Definition

- **Model Type:** 2D Pin-Jointed Truss
- **Analysis Method:** 2D Direct Stiffness Method (DSM)
- **Geometric Data:**
  - **Total Joints:** 34
  - **Total Members:** 75

2. Material and Section Properties

Uniform properties were assumed for all 75 members.

Property	Symbol	Value
Modulus of Elasticity	E	200 GPa (200×10^9 N/m²)
Material Density	ρ	7850 kg/m³
Cross-Sectional Area	A	0.01 m²

3. Boundary Conditions (Supports)

The structure is supported by **four pin supports**, which restrain translation in both the **X** and **Y** directions.

Support Type	Joint Numbers	Restrained Directions
Pin Support	1, 17, 33, 34	X and Y

4. Applied Loading

All loads were applied as **point loads at the joints**, and **g = 9.81 m/s²** was used for gravitational acceleration.

4.1 Dead Loads (DL)

1. **Member Self-Weight:**
   - Computed as:
     W=ρ×A×L×gW = \rho \times A \times L \times gW=ρ×A×L×g
   - The weight of each member was distributed as **lumped point loads**, with **50% of the weight** applied to **each of its two connecting joints**.

2. **Deck Dead Load:**
   - A uniform downward vertical load of **10 kN (10,000 N)** was applied to all **bottom chord joints (Joints 1–17)**.

4.2 Live Load (LL)

- **Load Magnitude:** 50 kN (50,000 N) per joint (downward).
- **Applied At:** Top chord joints **24, 25, 26, 27, and 28**.

5. Key Analysis Assumptions

1. The structure behaves as a **2D truss** — all members are **pin-connected** and carry **only axial forces** (tension or compression).
2. The material is **linearly elastic**, obeying **Hooke's Law**.
3. **Small displacement theory** applies — deformations are assumed small enough that geometry changes during loading are neglected.
4. **Member buckling effects** are **not considered** in this analysis.

6. Methodology

The analysis of the bridge truss structure was performed using the **Matrix Structural Analysis (MSA)** approach, specifically employing the **Direct Stiffness Method (DSM)**. This method provides a systematic and computational framework for determining structural displacements, internal forces, and reactions. The following steps outline the procedure used in the project:

1. **Formation of Member Stiffness Matrix (k′):**
   Each truss member was analyzed individually to establish its local stiffness matrix based on material properties (E, A) and member length (L).
2. **Computation of Local Forces (q):**
   The relationship between local forces and local displacements was defined as:
   q=k′dq = k′dq=k′d
3. **Transformation to Global Coordinates:**
   The local stiffness matrix and displacements were transformed into the global coordinate system using the **transformation matrix (T)**.
   d=TD,Q=TTqd = TD, \quad Q = T^Tqd=TD,Q=TTq
4. **Assembly of Global Stiffness Matrix (K):**
   The transformed member stiffness matrices were assembled to form the **global stiffness matrix**, representing the entire truss system:
   K=TTk′TK = T^T k′ TK=TTk′T
5. **Calculation of Global Displacements (D):**
   Using the equilibrium equation:
   Q=KDQ = K DQ=KD
   the global displacements of the joints were determined under applied loads.
6. **Determination of Internal Forces:**
   Once displacements were obtained, internal member forces were computed using the transformed stiffness relations:
   q=k′TDq = k′ T Dq=k′TD
7. **Interpretation of Structural Behavior:**
   The resulting displacements and internal forces were analyzed to identify tension and compression members, evaluate load paths, and verify equilibrium conditions.

7. CODE FOR ANALYSIS

1. Geometry, Material and Input Data

```
clear; close all; clc
Joints = [
% paste joints as rows: id, x, y
];
Members = [
% paste members as rows: id, node_i, node_j
];
E = 200e9;
A = 0.01;
rho = 7850;
g = 9.81;
numJoints = size(Joints,1);
numMembers = size(Members,1);
numDOFs = 2*numJoints;
jointIDs = Joints(:,1);
X = Joints(:,2);
Y = Joints(:,3);
id2index = containers.Map(jointIDs,1:numJoints);
DOF_Map = reshape(1:numDOFs,2,numJoints)';
```

Explanation:
This block initializes the truss model.
All joint coordinates and member connectivity are defined.
Material properties (Young's modulus, cross-sectional area, and density) are provided for steel.
Each joint has two degrees of freedom — horizontal (X) and vertical (Y) translation.
A mapping matrix assigns global DOF numbers to each joint.

2. Boundary Conditions

```
Supports = [
1,1,1;
17,1,1;
33,1,1;
34,1,1
];
```

```
restrainedDOFs = [];
for i=1:size(Supports,1)
    jid = Supports(i,1);
    idx = id2index(jid);
    if Supports(i,2)==1, restrainedDOFs=[restrainedDOFs;DOF_Map(idx,1)];
end
    if Supports(i,3)==1, restrainedDOFs=[restrainedDOFs;DOF_Map(idx,2)];
end
end
restrainedDOFs = unique(restrainedDOFs);
allDOFs = (1:numDOFs)';
freeDOFs = setdiff(allDOFs,restrainedDOFs);
```

Explanation:

This section applies the support conditions.

Pin supports are placed at four joints (1, 17, 33, and 34), restraining movement in both X and Y directions.

All restrained and free degrees of freedom are identified for later use in the stiffness matrix solution.

3. Stiffness Matrix Formation

```
K_global = zeros(numDOFs,numDOFs);
Loads = zeros(numDOFs,1);
MemberLengths = zeros(numMembers,1);
c = zeros(numMembers,1);
s = zeros(numMembers,1);
for e=1:numMembers
    n1 = Members(e,2); n2 = Members(e,3);
    i1 = id2index(n1); i2 = id2index(n2);
    x1 = X(i1); y1 = Y(i1);
    x2 = X(i2); y2 = Y(i2);
    L = sqrt((x2-x1)^2 + (y2-y1)^2);
    MemberLengths(e)=L;
    c(e)=(x2-x1)/L;
    s(e)=(y2-y1)/L;
end
for e=1:numMembers
    n1 = Members(e,2); n2 = Members(e,3);
    i1 = id2index(n1); i2 = id2index(n2);
```

```
    dofs=[DOF_Map(i1,1) DOF_Map(i1,2) DOF_Map(i2,1) DOF_Map(i2,2)];
    L=MemberLengths(e); ce=c(e); se=s(e);
    k=(E*A/L)*[ce*ce ce*se -ce*ce -ce*se; ce*se se*se -ce*se -se*se; -
ce*ce -ce*se ce*ce ce*se; -ce*se -se*se ce*se se*se];
    K_global(dofs,dofs)=K_global(dofs,dofs)+k;
end
```

Explanation:
This part constructs the **global stiffness matrix** using the Direct Stiffness Method.
Each truss member's local stiffness matrix is calculated based on geometry and material properties and then assembled into the global matrix that represents the entire bridge.

4. Load Application

```
MemberWeight=rho*A.*MemberLengths*g;
for e=1:numMembers
    J1=Members(e,2); J2=Members(e,3);
    i1=id2index(J1); i2=id2index(J2);
    w=MemberWeight(e)/2;
    Loads(DOF_Map(i1,2))=Loads(DOF_Map(i1,2))-w;
    Loads(DOF_Map(i2,2))=Loads(DOF_Map(i2,2))-w;
end
deckLoad=10e3; bottomChordJoints=1:17;
for j=bottomChordJoints
    idx=id2index(j);
    Loads(DOF_Map(idx,2))=Loads(DOF_Map(idx,2))-deckLoad;
end
liveLoad=50e3; liveLoadJoints=[24,25,26,27,28];
for j=liveLoadJoints
    idx=id2index(j);
    Loads(DOF_Map(idx,2))=Loads(DOF_Map(idx,2))-liveLoad;
end
```

Explanation:
This section applies **dead and live loads**.
Dead load includes self-weight of members (distributed to connected joints) and a constant deck load on bottom chord joints.
Live load simulates vehicular loads on top chord joints.

5. Solving for Displacements and Internal Forces

```
U=zeros(numDOFs,1);
U(freeDOFs)=K_global(freeDOFs,freeDOFs)\Loads(freeDOFs);
MemberAxial=zeros(numMembers,1);
for e=1:numMembers
    n1=Members(e,2); n2=Members(e,3);
    i1=id2index(n1); i2=id2index(n2);
    dofs=[DOF_Map(i1,1) DOF_Map(i1,2) DOF_Map(i2,1) DOF_Map(i2,2)];
    L=MemberLengths(e); ce=c(e); se=s(e);
    T=[-ce -se ce se]; ue=U(dofs);
    MemberAxial(e)=(E*A/L)*T*ue;
end
Reactions=K_global*U-Loads;
```

Explanation:
Using the reduced stiffness matrix, joint displacements are computed.
Member axial forces (tension or compression) are then calculated based on these displacements.
Finally, reaction forces at supports are obtained by multiplying the stiffness matrix with displacements and subtracting the applied loads.

6. Visualization

```
scale = 0.05*(max(max(Joints(:,2:3))) - min(min(Joints(:,2:3))));
Deformed = [X Y];
for j = 1:numJoints
    Deformed(j,1) = Deformed(j,1) + scale * U(DOF_Map(j,1));
    Deformed(j,2) = Deformed(j,2) + scale * U(DOF_Map(j,2));
end

figure('Name','Truss Deformation','Color','w');
hold on; axis equal; grid on;

for e = 1:numMembers
    J1 = Members(e,2); J2 = Members(e,3);
```

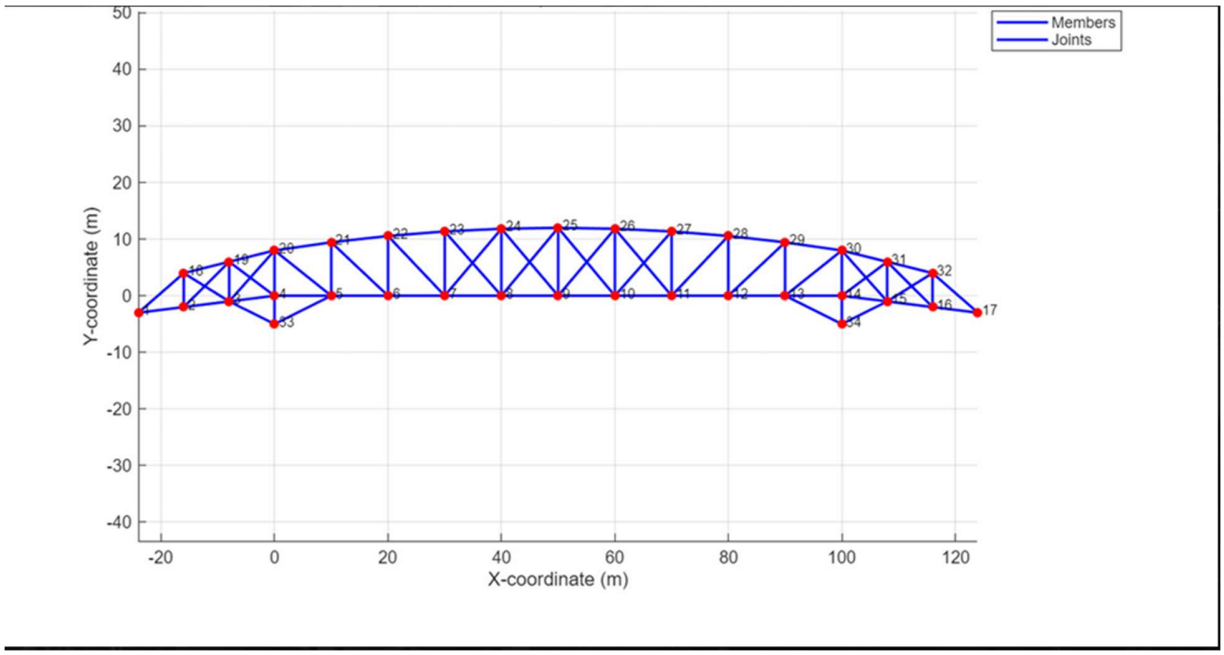
```
    i1 = id2index(J1); i2 = id2index(J2);
    plot([X(i1) X(i2)], [Y(i1) Y(i2)], 'k-', 'LineWidth', 1);
end

for e = 1:numMembers
    J1 = Members(e,2); J2 = Members(e,3);
    i1 = id2index(J1); i2 = id2index(J2);
    plot([Deformed(i1,1) Deformed(i2,1)], [Deformed(i1,2) Deformed(i2,2)],
'r-', 'LineWidth', 1.5);
end

supportX = X(cell2mat(values(id2index, num2cell(Supports(:,1)))));
supportY = Y(cell2mat(values(id2index, num2cell(Supports(:,1)))));
plot(supportX, supportY, 'bs', 'MarkerFaceColor', 'b', 'MarkerSize', 6);

plot(X(cell2mat(values(id2index, num2cell(liveLoadJoints)))), ...
     Y(cell2mat(values(id2index, num2cell(liveLoadJoints)))), ...
     'go', 'MarkerFaceColor', 'g', 'MarkerSize', 5);

legend({'Undeformed','Deformed (scaled)','Supports','Live Load Joints'},
...
       'Location','best');

xlabel('X (m)');
ylabel('Y (m)');
title('Truss Deformation (Scaled)');
disp('Member axial forces (N):'); disp(MemberAxial)
disp('Support reactions (N):'); disp(Reactions(restrainedDOFs))
```

Explanation

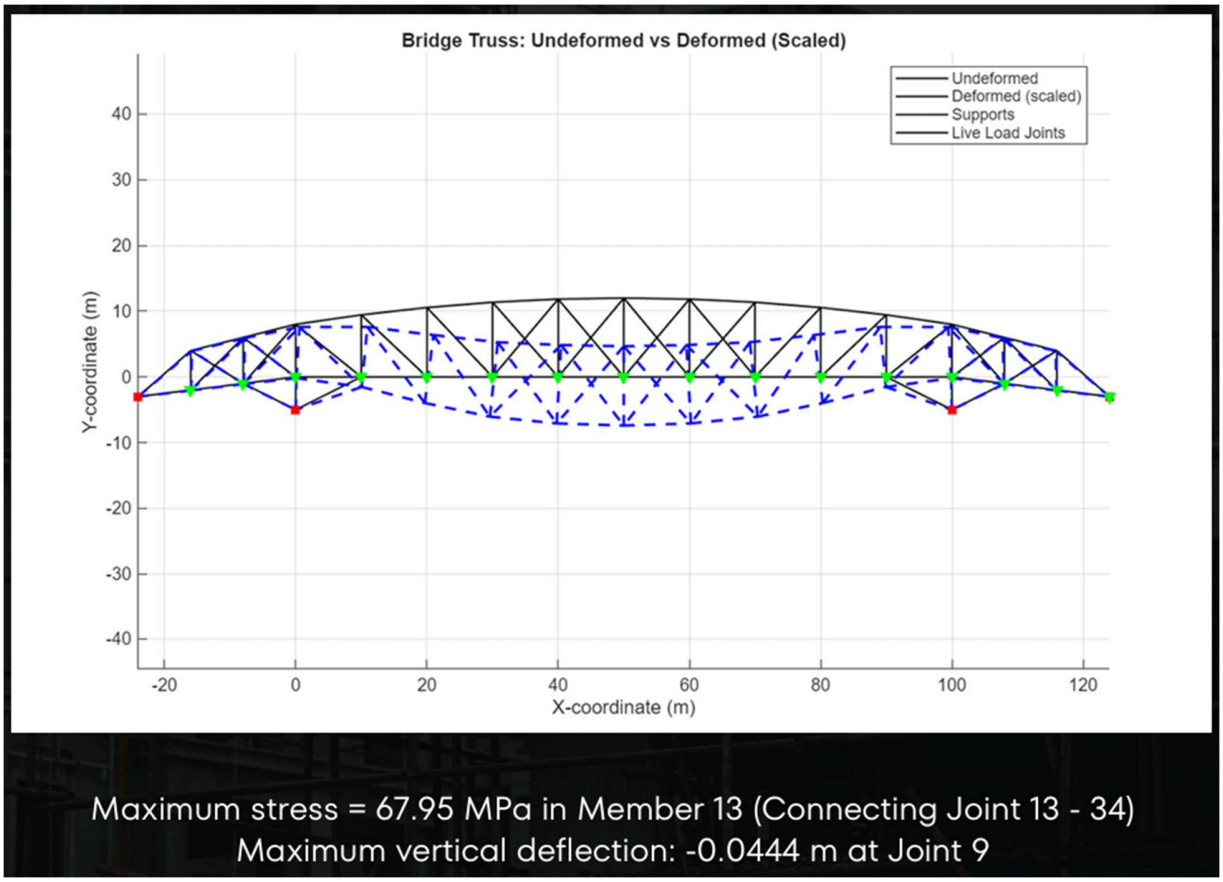
This visualization part plots the truss structure after analysis. The undeformed truss is shown using black lines, while the deformed shape is displayed using red lines to clearly illustrate the displacement of the structure under loading.The supports are represented by blue squares, indicating the fixed joints, and the live load joints are marked with green circles, showing where external loads are applied. A grid and equal axis scaling are used to maintain correct proportions and clarity in the graphical output.The plot thus provides a clear comparison between the original and deflected shapes of the truss, helping visualize how the structure responds to the applied forces.

8. RESULTS

Initial Structure:



Structural Deflection under Dead load + LIve load:



References

1) **Historical Context: General background information and the history of the Schell memorial bridge were sourced from Wikipedia.**
2) **Case Study and Theory: Detailed theoretical concepts, historical photographs, and specific details on the bridge's failure were drawn from the case study, "Schell memorial bridge" by Spaulding, M. & Ross, S. (2007)**
3) **Code implementation results are done in MATLAB.**
4) **Hibbeler, R. C. (2017). *Structural Analysis* (10th ed.). Pearson Education.**
5) **Nodal coordinates, member connectivity data, and visual aids (including diagrams of the idealized model and applied loads) were generated and exported from STAAD Pro.**
6) **https://www.canva.com/design/DAG3Eb0kXRM/bbssXY4nBElbXplHHD8FTg/edit?utm_content=DAG3Eb0kXRM&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton**